Vinayak Kamat

ML Engineer | Computer Vision, Evaluation & Data Pipelines

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Summary

ML Engineer with hands-on experience across the end-to-end vision workflow dataset creation, model training, ONNX optimization for deployment, and continuous evaluation. Built detection models in PyTorch (ulcer detection ~85% accuracy, eye-posture detection) and evaluation pipelines that gate releases on measurable quality. Comfortable reading and implementing papers, and working across backend, data, and product surfaces in a small team.

Skills

ML & Computer Vision: PyTorch, TensorFlow, Transformers, OpenCV, Object Detection, Model Training, Losses, Generalisation, Evaluation, Statistics

Data & Evaluation Pipelines: Dataset Creation, Data Curation, Benchmarking, Graders, Continuous Evaluation, Pandas, NumPy

Deployment & Optimization: ONNX, Model Optimization for Deployment, Docker, AWS (Lambda, S3, ECR, DynamoDB), GPU Workloads

LLM & Agents: RAG, LangChain, LlamaIndex, Prompt Engineering, Agentic Workflows, Chroma

Backend & Frontend: Python, FastAPI, Flask, REST APIs, React (basics), Git, SQL

Tooling: Claude Code / Cursor (agentic coding), MLflow, Experiment Tracking

Experience

KAN Innovations, ML Engineer — Mumbai, Maharashtra — May 2025 – present — 1 year 6 months

- Built and optimized vision models in PyTorch, then converted them to ONNX for faster on-prem/mobile inference, balancing accuracy against deployment latency
- Built data and evaluation pipelines for model releases, defining quality checks and benchmarks so changes were measured before shipping
- Built Vidur, a data pipeline that serializes results to JSON and uploads to S3 asynchronously so cloud transfer does not block execution
- Deployed ML services on AWS (Lambda, ECR, DynamoDB) and used MLflow for experiment tracking, model versioning, and reproducible workflows

KAN Innovations, ML Engineer Intern — Mumbai, Maharashtra — Feb 2025 – May 2025 — 3 months

- Built a PyTorch-based ulcer detection model achieving ~85% accuracy, from dataset preparation through training and evaluation
- Improved an existing eye-posture detection model through data and training refinements, measuring gains on held-out footage

Projects

Ulcer Detection — End-to-End Vision Model

PyTorch detection model trained, evaluated, and optimized for deployment.

- Prepared and curated training data, trained the model, and evaluated on held-out sets (~85% accuracy)
- Converted to ONNX to reduce inference latency for deployment on constrained hardware

Aethelgard — Evaluation & Guardrails for LLM Outputs
Real-time evaluation gateway that grades model outputs and flags risk.

- Built a FastAPI gateway using a secondary model to grade outputs for semantic risk and hallucinations
- Streamed evaluation metrics to Datadog for continuous monitoring, with fallbacks so low-confidence outputs degrade safely

nanoMind — On-Device Inference Optimization
Offline LLM app demonstrating low-latency inference on constrained hardware.

- Ran GGUF models on Android via llama.cpp with ARM-specific optimizations and real-time token streaming

Education

MSc **WorldQuant University**, Financial Engineering — Remote, Washington DC
Sept 2025 – May 2027

B.Tech **Punjab Technical University**, Computer Science — Punjab, India
Sept 2021 – May 2025

Achievements

Qualified GATE (Graduate Aptitude Test in Engineering)

Active open-source contributor with ML projects on GitHub